Summary post

by Abdulla Husain Salem Hadna Almessabi - Monday, 11 August 2025, 10:10 AM

The shift to Industry 5.0 leads to the focus on the balance of a high level of technology and human-friendly solutions, considering the resilience, sustainability, and ethical direction (Metcalf, 2024). This transition is vital in aviation and banking, in which safety and dependability are paramount. The CrowdStrike IT outage of 2024 affecting Delta Air Lines is a prominent instance of how interconnected systems are vulnerable (Alzaabi, 2024). This event highlights the need to implement distributed, redundant systems that are under human control based on Industry 5.0 concepts.

The TSB Bank IT migration failure in 2023 led to the interference with the ability of the customers to access them and immense financial losses (Habib et al., 2025). Such risks could have been reduced by the human-centeredness of Industry 5.0, as most of the risks can be prevented by contingency planning in terms of technology shifts (Kang & Cheung, 2024).

The Nissan ransomware in 2017 entailed operational and financial losses due to vulnerability in the Industry 4.0 environment. Since the manufacturing industry is dependent on IoT and automation, Industry 5.0 focuses on resilient systems that can be used to mitigate these risks (Vatsyayan et al., 2022).

These challenges can be overcome by using ABS, which aims at assisting organisations to model complex behaviours, automate decisions, and provide efficiency in decentralised settings (Ding et al., 2023). ABS has achieved flexibility and scalability and has created risks in terms of coordination and accountability, which can be handled by having governance structures and by integrating AI and IoT to enhance adaptability (Zhang et al., 2021).

Industry 5.0, which operates ABS as an enabler, offers a possibility of the development of resilient, sustainable, and human-centred systems, thus ensuring overall performance stability and consumer trust, and reducing technological risks.

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